

SAYAN DAS

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Summary

Software/AI-ML Engineer with **1+ year** of experience designing and shipping **scalable backend systems** and **production AI/ML pipelines**. Skilled in **LLMs, RAG, Agentic AI, System Design, FastAPI, REST APIs, GCP/AWS**. Published ML/NLP researcher with 2 conference papers. Delivered measurable impact – **99% performance gains** and **90% customer growth** on production systems. Targeting **AI/ML Engineer** and **Backend SDE** roles.

Technical Skills

Languages: Python, C, C++, JavaScript

Backend & System Design: FastAPI, Flask, Django, System Design, Testing (Pytest)

AI/ML & LLM: Machine Learning, Deep Learning, NLP, Transformers, RAG Pipelines, LLMs, LangChain, LangGraph, Agentic AI, Hugging Face

Databases: MySQL, PostgreSQL (Supabase, Neon DB), Firebase

DevOps & Cloud: Docker, GCP, AWS, CI/CD, Git, Linux

Other: Data Structures & Algorithms, n8n, Blockchain (Solidity)

Experience

Founding AI/ML Engineer

Jun 2025 – Present

f-ai

Bengaluru, Karnataka

- Designed and architected a **production AI/ML backend from scratch** (Python, FastAPI) following **scalable system design principles**, improving processing performance by **~99%** and driving **90% growth** in business customers.
- Built scalable **data schemas and extraction pipelines** enabling **~100% accurate** document processing and validation across diverse input formats.
- Pretrained, fine-tuned, and deployed **custom AI models** on domain-specific datasets using LangChain and LangGraph, with continuous evaluation and optimization loops.
- Designed and implemented **Agentic AI workflows** using n8n to automate internal operations, boosting team productivity.
- Built and deployed a **FastAPI-based microservice/monolith** for efficient REST API orchestration and low-latency model inference at scale.
- Established **CI/CD pipelines** enabling zero-downtime deployments across **GCP and AWS**.

Teaching Assistant – DBMS, DSA, Programming Paradigms

Aug 2023 – Jul 2025

IIIT Trichy

Tiruchirappalli, Tamil Nadu

- Led tutorials in DBMS, Data Structures & Algorithms, and Programming Paradigms for B.Tech CS students; graded assignments and mentored students, improving academic performance.

Research Intern

Jul 2022 – Aug 2022

Tripura University

Tripura

- Analyzed **blockchain/decentralized ledger** technology for vehicle anti-theft systems; identified 3 data-encryption gaps and proposed security enhancements.

Projects

Krishi Sahayta | Python, Flask, ML, NLP, LLM, React Native | [GitHub](#)

2022 – 2025

- Built a web chatbot and Android app for farmers (crop insights, market prices, buyer connections), integrating custom ML/NLP models with LLMs to achieve **~99% accuracy**.
- Finalist, Code Titans Hackathon** (Microsoft Azure, 2022).

Securing Cognitive Radio Network using Blockchain | Solidity, Ganache | [GitHub](#)

2023

- Built a Solidity-based blockchain model securing cognitive radio networks against SSDF attacks; validated resilience via Ganache simulation.

Publications

- Sayan Das, Dr. M. Ambika, “From Data to Harvest: Leveraging ML and Dataset Visualization for Crop Prediction,” *MIND 2024* (Springer CCIS, Dec 2024).
- Sayan Das, Dr. M. Ambika, “Revolutionizing Indian Farming: ML-Powered NLP for Optimal Crop Recommendations,” *ICLMDE* (Elsevier, Nov 2024).

Education

IIIT Trichy

2023 – 2025

M.Tech, Computer Science & Engineering — CGPA: 8.7/10

Tiruchirappalli, Tamil Nadu

NERIST

2019 – 2023

B.Tech, Computer Science & Engineering — CGPA: 8.52/10

Arunachal Pradesh

Profiles & Achievements

Competitive: LeetCode | HackerRank | Kaggle **Research:** Google Scholar | Hugging Face **Hackathon:** Finalist, Code Titans (Microsoft Azure, 2022)